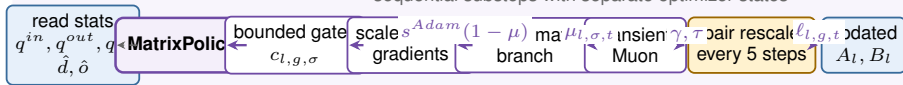


One optimizer step: two separated parameter lanes

RLB matrix lane: $M_{l,\sigma} \in \{A_l, B_l\}$ receives MatrixPolicy actions

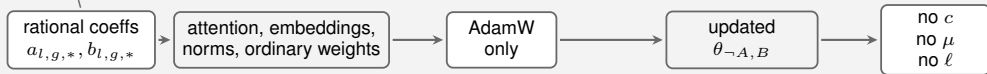
sequential substeps with separate optimizer states



$$A_g \leftarrow e^\ell A_g, \quad B_g \leftarrow e^{-\ell} B_g$$

q^R observed

AdamW-only lane: no MatrixPolicy gates, no Muon branch, no pair rescale



solid purple arrows stay in the matrix lane; dashed grey arrows are observations only